

Project Workflow - A Proposal

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Lab 02 Group E

Proposed Project Sequence

The following is a proposed sequence of steps for completing the Statement of Work (SOW).

1. Understanding the Project Requirements

1. **Read the SOW Document:** Understand the objectives, generate key questions, obtain answer and define deliverables required.
2. **Identify Key Stakeholders:** Fulton Hogan, Baa Ltd, and internal team members.

2. Set Up the Environment

1. Each member install software:

- Install R and RStudio.
- Install Git for version control.
- Install required R packages (e.g., `dplyr`, `sf`, `leaflet`).

2. Create Project Directories:

- Previously suggested directories for code (`code`), data (`data`), and documentation (`assignment`).

3. Create the GitHub Repository

1. Set Up a GitHub Repository:

- Create a new repository with the required naming conventions (e.g., DATA422\ Lab 02 Group E).

2. Upload Code and Data:

- Commit all R scripts, data files, and supporting documents to the repository.

3. Document the Repository:

- Write a `README.md` file describing the purpose of the project, the datasets used, and instructions for running the code.

4. Set Up and Use Trello for Project Management

1. Create a Trello Board:

- Set up columns for To Do, In Progress, and Done.

2. Assign Tasks:

- Create cards for each task (e.g., data cleaning, report writing) and assign them to team members.

3. Track Progress:

- Update cards with comments, and move them between columns as tasks are completed.

5. Data Acquisition

1. Obtain Data Files:

- Download and keep a local copy of the data files listed in the SOW (e.g., `sp_data.csv.gz`, `vf_data.parquet` from the zip file).

2. Verify Data Integrity:

- Ensure that all files are complete and free from corruption.

6. Data Cleaning and Preprocessing

1. Load the Data into R:

- Use R scripts to read in CSV and Parquet files.

2. Clean the Data:

- Handle missing values, remove duplicates, and standardise formats (e.g., dates).

3. Merge and Transform Data:

- Use concordance files to merge datasets, creating a unified dataset with relevant variables.

4. Generate Additional Variables:

- Calculate metrics such as population density or average daily traffic volumes as required.

7. Exploratory Data Analysis (EDA)

1. Visualise Traffic Patterns:

- Create charts to compare traffic and pedestrian volumes during school holidays and regular periods.

2. Identify Trends and Outliers:

- Use summary statistics and visualisations to understand data distributions and trends.

8. Answer Key Questions

1. Does it Make Sense to Plan Roadworks for School Holidays?

- Compare traffic and pedestrian activity between school holidays and regular periods.

2. Which Days are Best for Daytime Roadworks?

- Identify days with the least traffic congestion and business activity.

3. Are There Geographical Differences?

- Analyse differences in traffic and business activity between Auckland, Wellington, and Christchurch.

9. Generate the Cleaned Dataset

1. Create the Final Dataset:

- Output the cleaned and processed data as a single gzipped CSV file.
- Include Territorial Authority code, SA2 code, datetime in NZST, and a count of people in the area.

10. Create the Technical PDF Report

1. Write a Detailed Methods Section:

- Describe the data processing pipeline, including any transformations or calculations performed.

2. Highlight Limitations and Future Work:

- Discuss any limitations of the data and suggest areas for future research or improvements.

11. Create the Non-Technical PDF Report

1. Summarise Key Findings:

- Present the answers to the key questions in a clear and concise manner.

2. Include Visuals:

- Use charts and graphs to support findings and make them accessible to a non-technical audience.

12. Final Review and Submission

1. Review All Deliverables:

- Check the code, cleaned dataset, reports, and Trello board for completeness and accuracy.

2. Submit the GitHub Repository:

- Ensure the repository is public or shared with the client, and provide the link in the README.

3. Submit the Reports:

- Submit the technical and non-technical PDF reports as per the assignment guidelines.